

Market Microstructure and Electronic Trading Finance 480

Dale W.R. Rosenthal¹

University of Illinois at Chicago,
College of Business Administration

30 August 2016

¹daler@uic.edu

Orders and Quotes

Recall last lecture we discussed microstructure:

- What it is (and is not); and,
- Why microstructure and electronic trading go together.

We also discussed market macrostructures:

- OTC Markets;
- Auction Theory;
- Fixings and Auctions (esp. open and close auctions);
- Sporadic Call Markets; and,
- Continuous Trading.

Readings on Today's Topic: Orders and Quotes

Readings, in order of importance:

- Chapter 4 of Johnson.
- Chapter 1 of Foucault, Pagano, and Röell.
- Blackboard: Readings on short sales, etc.
- Pages 191–200 of O'Hara (on reserve).

The first two are required.

The last text is on reserve at Daley Library.

- Today we'll talk about why markets exist: want².
- Specifically, we'll talk about expressing want through *orders*.
- However, an order implies want and sacrificed flexibility.
- Some orders may be restricted or tagged for special handling.
- A non-ephemeral order may be quoted.
- Thus *quotes* result from (some) orders.

²Or perhaps *potrzebie*.

Orders

- Expressing want: we can be attracted, repelled, or indifferent.
- Thus orders can be to buy or sell (or can not exist).
- Regulations in many countries constrain *short sales*.
- May even need to specify if a buy covers a short position.
- Thus there are more than just two types of orders.

- In US equity markets, there are four basic order types:
 - Buy (B);
 - Sell (S);
 - Sell Short (SS); and,
 - Sell Short Exempt³ (SSE).
- SSE's are used by official market makers to sell short:
 - without borrowing the security; and,
 - (erstwhile) without being subject to a price test.
 - Thus they exist to increase liquidity.

³Under Reg SHO, SSE's are merely markings.

How Short Sales Work

- Selling short first requires borrowing securities from somebody.
- Most firms require that you get *locates*.
- You then sell those securities.
- Later, you buy securities back and return them to the lender.
- The recent short selling brouhaha was about this.
 - Some traders shorted “naked” (without locates).
 - They then covered their shorts before delivery.
 - SEC required some shorts (banks) to have unique locates.

Selling Short: Controversy

Short selling is often controversial.

- Profit when a security/market declines in value.
- Was seen as unpatriotic for much of the 20th century.
- Has often been restricted, even in the US.
 - AMEX/NYSE subjected short sales to a “0+ tick” rule.
 - Nasdaq subjected short sales to a “bid-test” rule.
- Even now: inconsistent attitude to shorts is stunning.
 - We blame shorts for bringing down bank stocks.
 - Yet “oil prices are high because of speculators.”
 - Who is thanking speculators for falling oil prices?

Short Selling: Reg SHO

- Pilot: 3 Jan 2005; Live: 3 Jul 2007.
- Affects short sales in all US securities:
 - Suspends SEC and SRO⁴ price tests restricting short sales.
- You must:
 - “Believe the security can be borrowed” to meet delivery; and,
 - Document the locate before shorting.
- SEC creates “threshold” list of securities with excessive fails.
- Broker-dealers⁵ must:
 - Close-out threshold security fails after 13 settlement days; and,
 - Not short those securities until such close-out is complete.

⁴Self-regulatory organizations, like Finra.

⁵Option MM exception.

Selling Short: Unstoppable

Short selling is beneficial:

- Short selling rewards investors for exposing problems.
- That oversight guides market prices toward fair value⁶.

You cannot stop short selling:

- Short exposure can also be had via swaps or options.
- For indexed securities, you can synthetically short:
 - Do an EFP of an index futures contract.
 - Short an index futures contract.
 - Sell (long) the securities you want to short.
- *N.B.* Also useful if you cannot get dependable borrow.

⁶Representative prices attract more investors.

Price Immediacy

Orders may trade immediacy for price sensitivity.

- *Market* orders execute immediately against “far side.”
- *Limit* orders specify a worst acceptable price.
 - *Marketable limit* orders should execute immediately.
- *Market if touched* (MIT) orders activate at a given near price.
- *Stop* (loss) orders activate at a given far price.

MIT buys activate below current market; stops activate above.

One perspective on margined positions and margin calls:

- implicit stop orders which can be canceled at activation.

Non-market orders may exist for a specified time period.

- *Good-til-canceled* (GTC) orders stay active⁷ until canceled.
- *Day* orders are canceled after the close/closing auction.
- *Immediate or cancel*⁸ (IOC) and NOW orders:
 - canceled after trading against book/CLOB_{fast}.

And for US equities:

- *Extended hours* (X, GTX, IOX) orders are also active in:
 - the pre-market (8:00–9:15) and
 - the after-market (16:15–18:30).

⁷Some brokers limit how long GTCs live.

⁸LSE calls these execute or eliminate (EOE) orders.

Open/Close Orders I

Open and close orders are slightly different:

- Market orders may trade at any price:
 - *Market on open* (MOO/OPG) orders trade at the open⁹.
 - *Market on close* (MOC) orders trade at the close.
- Open/close limit orders may trade at limit price or better.
 - *Limit on open* (LOO) orders trade at the open.
 - *Limit on close* (LOC) orders trade at the close¹⁰.
- Imbalance only (IO) orders¹¹ trade against auction imbalances.
- *Funari*: a TSE limit order which becomes an MOC at close.

⁹Market orders sent pre-open will MOO.

¹⁰Limit orders sent immediately pre-close approximate LOCs.

¹¹Only available in Nasdaq open/close.

Other details about open and close orders:

- Even market orders may not fully trade; partial fills can occur.
- Futures/options: orders may trade in an open/close range.
- Equities: orders usually trade in an auction.
- One exception: Eurex futures start/end via auction.

¹²E.B.

Amount and Price Qualifiers

Orders may constrain the acceptable amount to be filled:

- *All or none* (AON) orders cannot be partially filled.
- *Fill or kill* (FOK) orders are like AON IOCs.

Orders may also control price dynamics:

- *Pegged* orders base their limit price on the “near side.”
- *Reverse pegged* orders base their limit price on the “far side.”
- *Intermarket sweep* (ISO) orders trade through many prices.
 - SuperMontage only sweeps 10% + \$0.01 past the “near side.”

Orders may hide their true amount or limit price.

- *Reserve* (“iceberg”) orders show less quantity than available.
 - Refreshes to display size until total size traded.
 - Price-time priority for displayed amount.
 - Total amount available on trade-through.
- *Discretionary* orders will trade at a worse price than shown.
 - Do not get time priority;
 - Last in queue at a given price level.

Routing: used to send trades to venues with better prices.

- Only a problem in decentralized markets.
- US: Routing to better prices required to participate in NBBO.
- NBBO representation attracts liquidity and, thus, traffic.
- However, orders may also be *directed*.
- Some orders are not routed:
 - Pegged and reverse pegged orders;
 - Sweep orders, except intermarket sweeps (ISOs);
 - *Fill or return* (FOR) and *post only*¹³ orders; and,
 - IOC (versus NOW orders).

¹³ Absolutely, definitely not POO.

Routing: Flash Orders

- Can we get a “second try” at an order before routing away?
- One way: *flash* or *step up* orders:
 - Send order at NBBO/better to exchange clients;
 - Order times out after short time (e.g. 500 ms); and,
 - Route unfilled part of order away after timeout.
- Example: Buy order of 1,000 shares up to \$20.10.
 - Nasdaq offers 400 shares @ \$20.03, 1,000@\$20.05.
 - BATS offers 700 shares @ \$20.04.
 - Order sent to Nasdaq, 400 filled @ \$20.03.
 - Nasdaq flashes order to its clients as “B 600@\$20.04.”
 - After 500 ms, Nasdaq routes remaining order to BATS.
- Issues?

Another issue is the trader's responsibility:

- *Held* orders hold trader responsible for “best execution.”
- *Not held* orders give trader discretion to work the order.
- An *agency* order represents an order from a client.
 - “Agent” implies fiduciary duty.
 - Thus usually¹⁴ agency orders are held orders.
- A *principal* order is not a client order¹⁵.
 - Thus principal orders are “not held” orders.

¹⁴Except if for large orders.

¹⁵Except riskless principal deals.

Order handling rules are key in centralized and distributed markets.

- Usually must fill agency before agent's principal orders¹⁶.
- Rules may not apply to odd-lot or block-size orders.
- Distributed markets: protection may not apply across venues.
- OTC bond markets: order handling is even more nebulous.
- If an agent's principal orders step in front of agency orders:
 - this is "front running"; and,
 - the SEC takes this very seriously.
 - MiFID¹⁷ also lists this as a very serious offense.

Discrepancy: locates are not subject to agency protection¹⁸.

¹⁶Nasdaq's rules are known as the Manning rules.

¹⁷The EU Markets in Financial Instruments Directive.

¹⁸Harris fails to note this on p. 146.

Compound Orders

Orders may be combined:

- *Spread* orders buy and sell related securities¹⁹.
- *Repurchase* (repo) orders sell and agree to repurchase later²⁰.
- *Sell/buy* orders are sells coupled with forward buys.
- A repo is one trade; a sell/buy is two trades.

Orders may also be contingent:

- *One cancels other* (OCO) orders; and,
- *One sends other* (OSO) orders.
- Not all brokers or market centers accept OCOs and OSOs.

¹⁹ e.g. Crush spread.

²⁰ *Reverse repos* are the other sides of repos.

Some issues complicate thinking about orders:

- Changing an order (COR/CXR/CFO)²¹ may not be atomic.
 - May have to cancel, wait²², send new order.
- Market centers may pay for order flow.
 - Your PA trades are probably routed to a wholesaler²³.
- Some market centers rebate you for providing liquidity;
 - They may also charge for demanding liquidity.
- Limit orders are like writing very short-term options.

We will cover some of these issues later in the course.

²¹Correct; cancel-replace; cancel with following order.

²²Unless cancels are granted without adverse selection.

²³Payment for flow was once 25% of E-Trade's revenue.

Quotes

- Recall market orders do not persist; they trade, then go away.
- *Quotes* are collections of active limit orders.
 - Buy orders are collected by price and become *bids*;
 - Sell²⁴ orders are collected by price and become *offers*.
 - Offers are also known as *asks*.
- So quotes are prices at which people stand ready to buy/sell.

²⁴Including sell short. Why?

- The highest bid amongst a set is the *best bid*.
- The lowest offer amongst a set is the *best offer*.
- Together these are the *best bid and offer* (BBO).
- BBO across all market centers: national BBO (NBBO).
- NBBO is better known as the *spread* or the *inside market*.
- In the UK, the spread is often called the *touch*.

- If you buy/sell:
 - the best bid/ask is the *near side* of the spread; and,
 - the best ask/bid is the *far side* of the spread.
- An order trading at the far side is said to *cross the spread*.
 - A sell crossing the spread *hits* the bid(s).
 - A buy crossing the spread *lifts* or *takes* the offer(s).
- Trading ceases while spreads are zero or negative:
 - If best bid = best ask, the market is *locked*.
 - If best bid > best ask, the market is *crossed*.
 - Orders must trade or move before trading continues.

In distributed markets, information holds the market together.

- *Display Rules* for limit orders are thus important.
- Display rules may require display of order's price and size.
- Exceptions may be made, however.
- On Nasdaq:
 - Odd-lot orders are not displayed;
 - AON orders are not displayed; and,
 - block-size²⁵ orders are not displayed.
 - Worse: Display is required within 30s (an eternity).

Hence the concern about getting “pennied.”

²⁵Minimum 10,000 shares and \$100,000.

Display Rules: Weird Science

In July of 2004, I conducted a few experiments:

- Odd buy to below ask.
- Odd sell short to lock.

Also, recall the display instructions on orders.

Those instructions first existed to ease market makers' duties.

- Now those features affect information dispersal via the book.
- Reserve/iceberg orders display less amount than available.
 - Thus available quantity may be $>$ displayed quantity.
- Discretionary orders display less-aggressive price than available.
 - Thus available price may be more aggressive than displayed.

Quote Collusion?

What can we do with what we've learned so far?

Surprisingly, we can now understand a key microstructure paper.

- Christie and Schultz, "Why Do NASDAQ Market Makers Avoid Odd Eighth Quotes?" *Journal of Finance*, 1994.
 - Let's look at Figure 3 on page 1823.
 - Is the quoting pattern that unusual?
 - Try some tests...
-
- Outcome: SEC charged >40 market makers with collusion.
 - Resulted in settlements totalling more than \$1BB.

A *limit order book* (“book”):

- is the total amount at each price for standing limit orders.
- changes every time a limit order is:
 - received;
 - corrected;
 - canceled; or,
 - filled.
- often excludes non-“standard-way” orders.
- may show individual order amounts at each price level.

Limit Order Books: Issues

Some issues to consider when viewing order book data:

- Venues may only offer a partial view of the book.
- Venues may not offer an order book.
- Venues may not disseminate the book in a timely fashion.
 - NYSE OpenBook was delayed 15–30s for years.
 - This is more of a problem in specialist markets.
- Book storage/indexing may require significant computation.
- Book analytics may be gamed by flashing large orders.

Limit Order Books: Examples

Book names and features vary:

Venue	Book Name	Detail Offered
NYSE	OpenBook Ultra	Full Depth+
ArcaEx	ArcaBook	Full Depth
Nasdaq	Level 2/TotalView	Full Depth
LSE	Level 2	Full Depth
Euronext	Cash Full Order Book	Full Depth
Eurex	Level 2	Top N^{26} bids/asks
CME	Market Depth	Top 5 bids/asks
ICE	Level II	Full Depth
LIFFE	Commodity FOB	Full Depth
ISE	Depth of Market	Top 5 bids/asks
eSpeed	BGCantor Market Insight	Top 3 bids/asks

²⁶ N varies by product.

The Consolidated Limit Order Book (CLOB)

- Reg NMS: unite LOBs into a *consolidated* book (CLOB).
- The carrot:
 - Participate in the CLOB by offering inside quotes.
 - Route away to better prices.
 - Receive market-wide flow when you are best price.
- The stick:
 - Non-CLOB venues cannot disseminate quotes.
 - Routing to non-CLOB venues complicates “best execution.”
- Idea: CLOB eases “best execution” for all market participants.
- CLOB also eliminates annoying “trade-throughs.”

CLOB: Anti-competitive?

How could a CLOB reduce competition?

- If orders must be sent to CLOB:
 - Creates a single market with merely different entry points.
- If venues need not route to destination with best price:
 - Venues can selectively price improve.
- If venues must send quotes and route orders to one another:
 - Slow venues reduce CLOB to *slowest* common denominator.
 - Solution?: Allow orders to be marked as fast and firm (NOW-able).

Other problems with a CLOB:

- Large orders may want anonymity/hidden flexibility.
- CLOBs must handle/accept different clocks and latencies²⁷.
 - Price-time priority is very difficult to maintain.
- Price levels ignore (complicated) payment-for-flow rebates.
- Traders can make DIY CLOBS; why command-and-control?

²⁷ A word or two on *latency*.

CLOB versus Supply and Demand Curves

A CLOB is *not* excess supply+demand curves.

CLOBs do not include:

- Opportunistic traders who desire stealth.
- Large customer orders which are being traded across time.
- People who trade if large price changes happen too quickly.
- Orders which only trade at certain times.
- Orders for non-standard delivery and/or settlement.
- Investors who forget positions until some price range is hit.
- Traders who see showing an order as inviting adverse selection.

Remember: Posting limit orders is like writing free options.

The Road Ahead

We've talked about market macrostructure, orders, and quotes.
We are almost done with the basics:

- The Basics: Trades and Traders
- Market Structures: Continuous Markets; Decentralization
- Metrics/Models: Prices; Sizes; Limits; Time; Liquidity
- Electronic Trading: Markets; Tools; Tactics; Strategies
- Microstructure and Policy Issues

Homework 1 will be posted this week.

Readings, in order of importance:

- Chapter 2 of Johnson.
- Blackboard: Readings